

Augmented Reality Guidance for Numerical Control Program Setups

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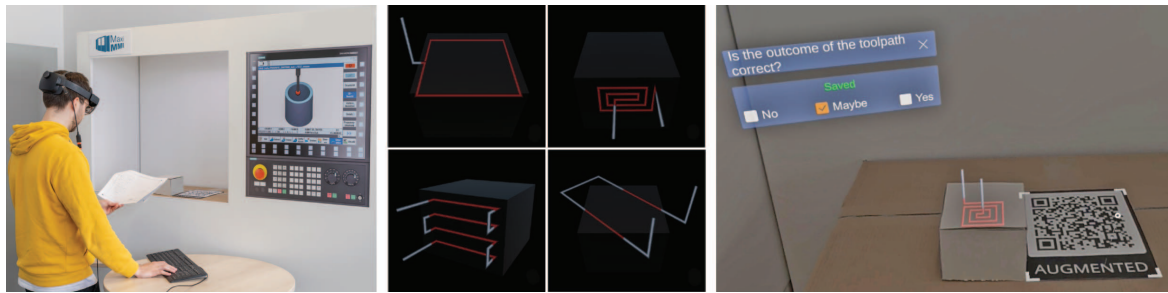


Figure 1: Our overlays a blank with the planned milling trajectory (examples center), enabling operators to visually assess correctness. In a user study, participants evaluated these visuals using conventional paper-based technical schematics (left). A virtual interface allowed participants to input whether a displayed path will yield the intended product (right, users perspective).

ABSTRACT

Setting up a new numerical control program is a difficult and tedious task that can cause costly errors. Diagnostic information is commonly presented as abstract numerical data. Collaboratively with machine operators, we designed, implemented and evaluated Augmented Reality guidance methods based on their insights. A visual milling path trajectory was chosen for further evaluation. Results indicate that this path preview can decrease error detection time, increase detection rate, and enhance user confidence compared to a typical 2D machine panel. Users quickly noticed errors in paths, while errors related to milling depth were slower to detect.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Interaction design—Interaction design process and methods—User centered design Applied computing—Operations research—Industry and manufacturing

1 INTRODUCTION

Automation has advanced modern industrial production, but some processes still need human-in-the-loop interactions. One such process is the setup. Computer numerical control (CNC) machine programs are usually created in a dedicated development environment [1]. Programs are then adapted and optimized for a new workpiece at the machine, using the often suboptimal and ergonomically poor built-in displays [2]. The operator has to inspect the tool path for errors, such as incorrect speed, cutting through air, or spindle collisions. These errors can affect the quality of the workpiece or damage the machine. Modern simulation methods can support this process, but they have limitations in accuracy, computation, and features [6]. Therefore, manual inspections are still necessary.

Information about the raw blank, final workpiece, manufacturing process, and milling path is regularly provided on paper. The opera-

tor runs the program incrementally, often at reduced speed, to check the accuracy. This can take a long time and be cognitively demanding, especially for complex products. To address these challenges, we propose Augmented Reality (AR) guidance visualizations, as shown in Fig. 1. Showing increasing adoption, AR is used throughout the industry to aid workers [4]. As user acceptance is key to successful AR support [5], our visualizations are based on expert interviews. Showing aligned information is more accessible and user-friendly than conventional 2D interfaces, and existing works explored showing machine parameters in place [7], using digital twins of the workpiece [3]. Thus, showing path trajectory information can not only help novice operators to set up machines effectively, but also speed up the setup process and improve early error detection. Using an AR head-mounted display (HMD) is the most practical way to achieve this, given the large number of existing CNC mills.

2 SETUP PROCESS ANALYSIS

We aimed to understand the current situation and challenges of CNC machine setups. We involved real, experienced machinists in our process. This grounded our work in real user problems and solutions. The focus was on a milling setup with three-axis machines and stationary workpieces, for simpler tool paths and visual guidance. We analyzed the setup process for a CNC mill in two steps. First, we explored the status quo, especially for novices, with expert interviews. We also collected and rated possible errors and their detection parameters. Especially parameters from the machine's panel are relevant, as they can be utilized in AR guidance. We categorized and ranked the errors by frequency, severity, damage, and AR value. A simplified risk matrix can be seen in Fig. 2 on the left. This revealed a high damage and cost potential in a substantial proportion of these errors, primarily due to the risk of tool or spindle damage. Values typically set empirically were not considered further in our error analysis. Of critical importance in detection were machine messages, alarms, and status indicators. The 3D vector, together with the G-function vector (indicating the tool's next programmed movement), was also of particular interest. It's to note that most operators also heavily relied on their experience and used non-machine parameters, like the sound of the machine and the cutting removing material, or the size, speed and shape of removed shavings.

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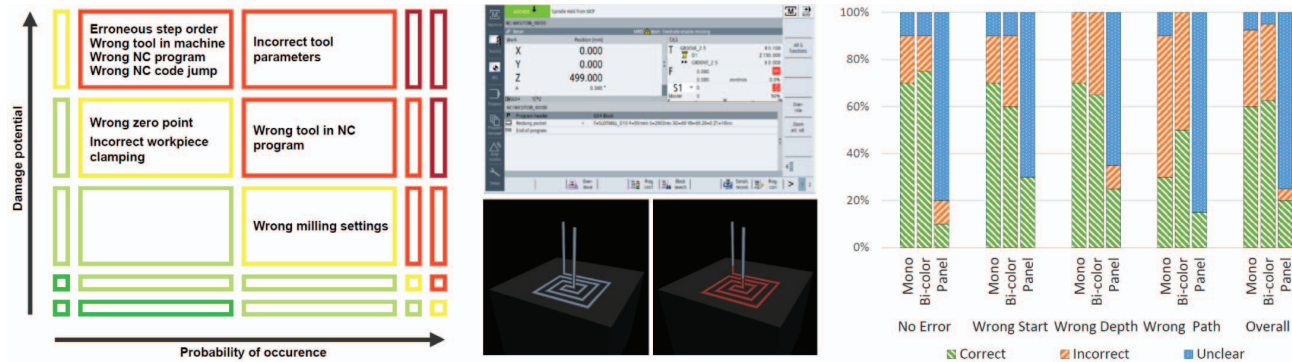


Figure 2: Risk matrix (left) created with expert interviews shows most errors have high damage potential. Middle shows the three visualization types used in the user study, right hand graph shows experiment results.

Based on these errors, we designed and discussed 2D mockups of visual guidance in a second step. We identified the best visualizations from the mockups together with experts. Visualizations like shadow, laser dot, axis, position, and grid were discarded. They lacked information, clarity, or spatial accuracy, as the blank's position and size were unknown in the setup process. This led us to focus on the visualization of a path preview for the machine as our most promising visual guidance strategy. Such a visualization would address multiple identified errors by previewing, for instance, where and how a thread is milled, or whether a hole or a pocket is oversized. In addition, this approach would enable the operator to spot errors like improperly clamped blanks, wrong zero point and inappropriately selected numerical control (NC) programs.

3 EXPERIMENTAL PROCEDURE AND RESULTS

A user study investigated how tool path visualization with an AR device can help users spot errors in the machine setup process, and compared it to the baseline. We visualized a CNC machine's next step, showing both correct and erroneous paths. Paths were shown in three different ways: machine interface (baseline), monochrome AR path, and bi-color AR path (see Fig. 2, middle). We tested five different paths, see Fig. 1 middle and left, and four error types: No error, wrong depth, wrong start, wrong paths. The paths were of realistic complexity for the machine and workpiece. Paths entered the workpiece from the top and sides. Participants evaluated the machine setup as correct, erroneous or unclear with an AR interface. The study was a within-subjects repeated measures design, we ensured that each participant only saw a specific path-error combination with one visualization type. The two dependent variables we measured were the responses to error detection (yes, no, uncertain), and the time taken to complete all assessments, in seconds. 12 participants (9m, 3f), employed at a company utilizing CNC machines, took part in the study. They had varying levels of familiarity with AR and NC machines. The setup consisted of a CNC replica, with a $12 \times 12 \times 9$ cm cardboard blank inside, and a HoloLens as display. Alignment of paths happened via a visual marker placed next to the workpiece. Participants had a short acclimatization phase.

The panel visualization elicited a high number of uncertain responses (75%), compared to 7.5% for the monochrome and 5% for the bi-color visualizations (see Fig. 2, right). Participants delivered accurate judgments 60% of the time for the monochrome, 62.5% for the bi-color and 20% of the time for the panel visualization. Statistical analysis using a one-way repeated measures ANOVA revealed a significant difference between panel ($M = .16$, $SD = .07$) and monochrome visualization ($M = .56$, $SD = .06$), $p = .003$, as well as bi-color visualization ($M = .61$, $SD = 0.05$), $p < .001$. There was no significant difference between monochrome

and bi-color visualizations, $p = 1.0$. The different visualization strategies did not yield any significant shifts in time spent on decisions, $F(2, 22) = .664$, $p = .525$, $\eta_p^2 = .057$, depending more on error type instead. Test further revealed significantly higher head movement in the path visualization conditions compared to the panel.

4 DISCUSSION

Our survey revealed high error potential, motivating the need for better guidance. AR visualizations can provide such guidance with appropriate, accurate tracking, with interviewees citing better comprehension and anomaly detection. Trajectory visualization boosted participants decision-making confidence, but also lead them to make more errors, possibly due to the approachable visualization compared to the panel. Participants identified more errors correctly, however mainly focused on offset and depth, missing alterations in the path. Care must be taken to not instill undue confidence in novices with such systems. We did not see a difference between monochrome and bi-color visualization, suggesting the HoloLens provides sufficient depth information. Still, with more complex situations like carving from behind, colorization might be helpful.

While we used small workpieces, a machine replica and a simple study setup, limiting applicability, we didn't observe any issues threatening data validity. Further works could improve on this with a more realistic setup, more participants, evaluation of more complex visualizations and impact of tracking quality.

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